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Yen

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(54) **INDUCTOR DEVICE**

USPC 336/200, 232, 225
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 846 days.

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English translation of TWI692780 (Year: 2020).*

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(30) **Foreign Application Priority Data**

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H01F 17/00 (2006.01)

H01F 27/29 (2006.01)

(52) **U.S. Cl.**

CPC **H01F 27/29** (2013.01); **H01F 17/0013** (2013.01); **H01F 2017/0073** (2013.01)

(58) **Field of Classification Search**

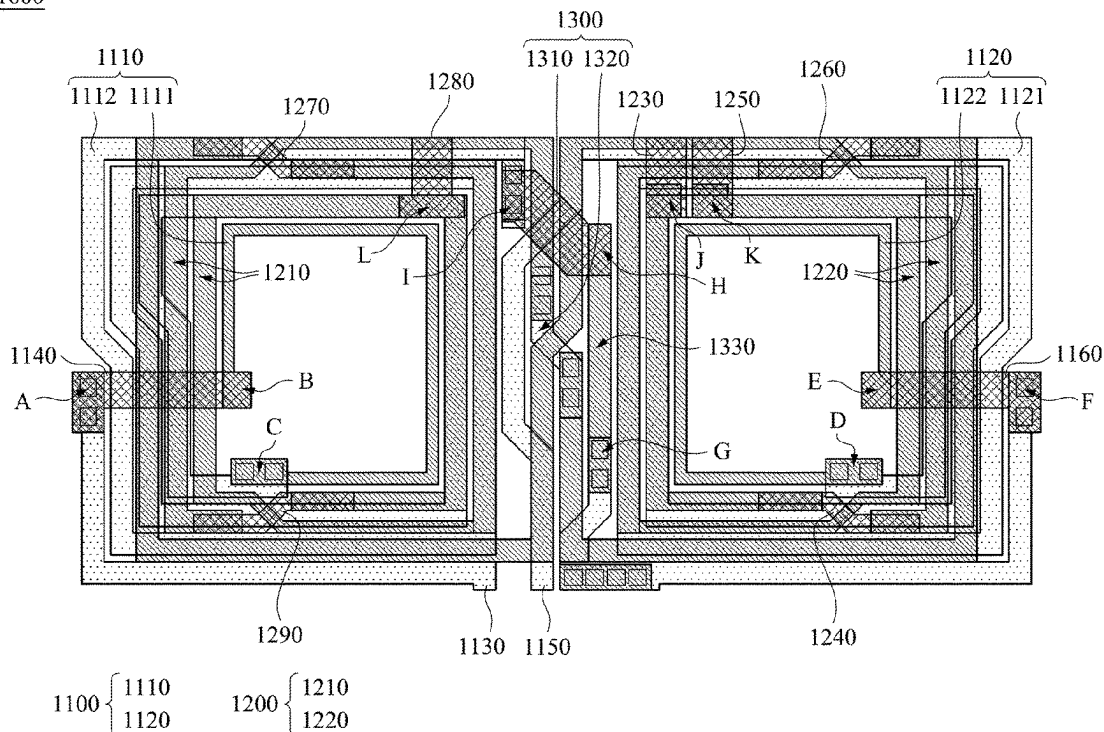
CPC H01F 27/29

(57) **ABSTRACT**

An inductor device includes a first trace, a second trace, and a connection member. The first inductor includes a first trace and a second trace. The shape of one of the first trace and the second trace is spiral. The second inductor includes a third trace and a fourth trace. The connection member is configured to couple the first inductor and the second inductor.

17 Claims, 8 Drawing Sheets

1000



1000

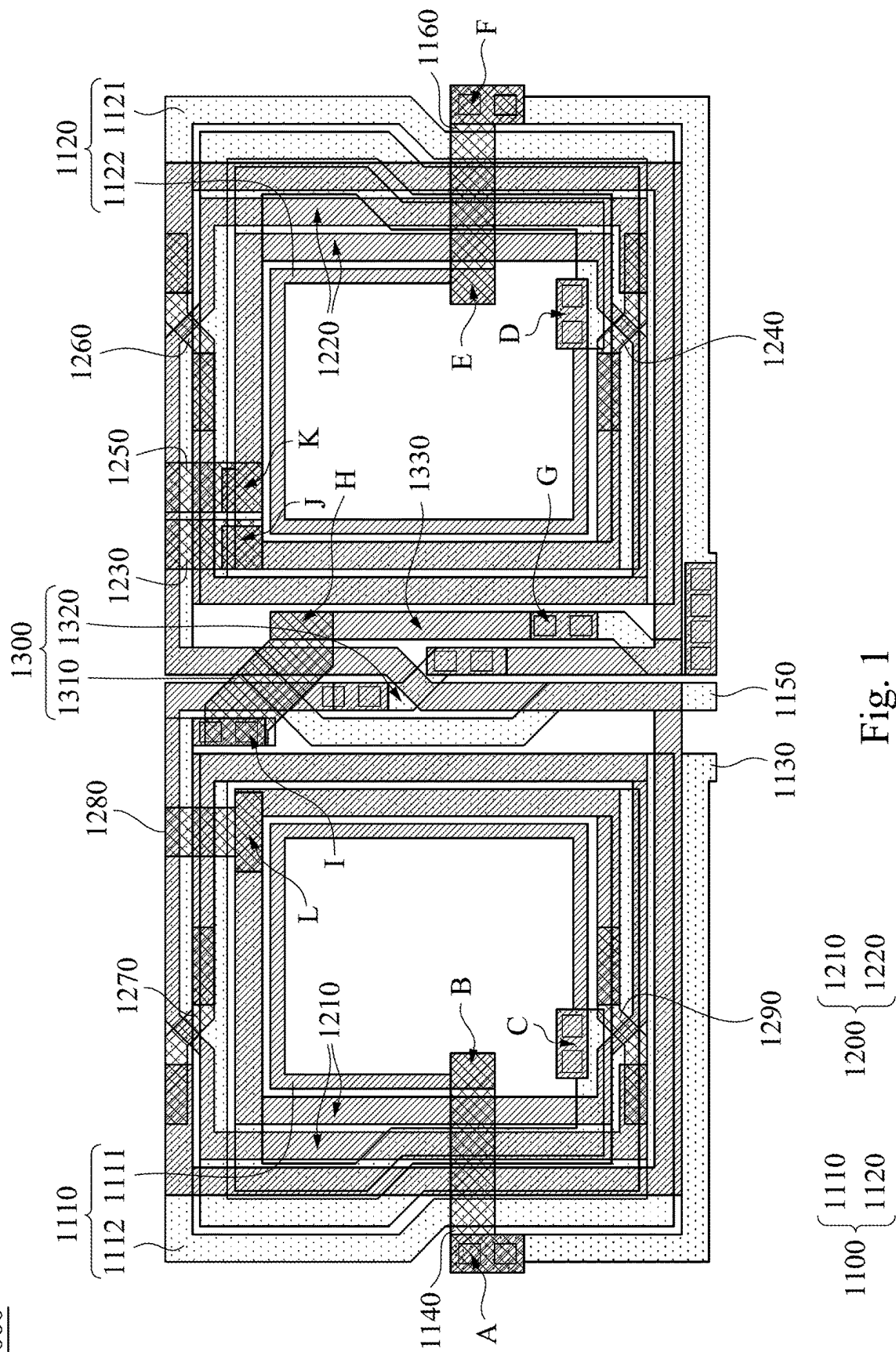


Fig. 1

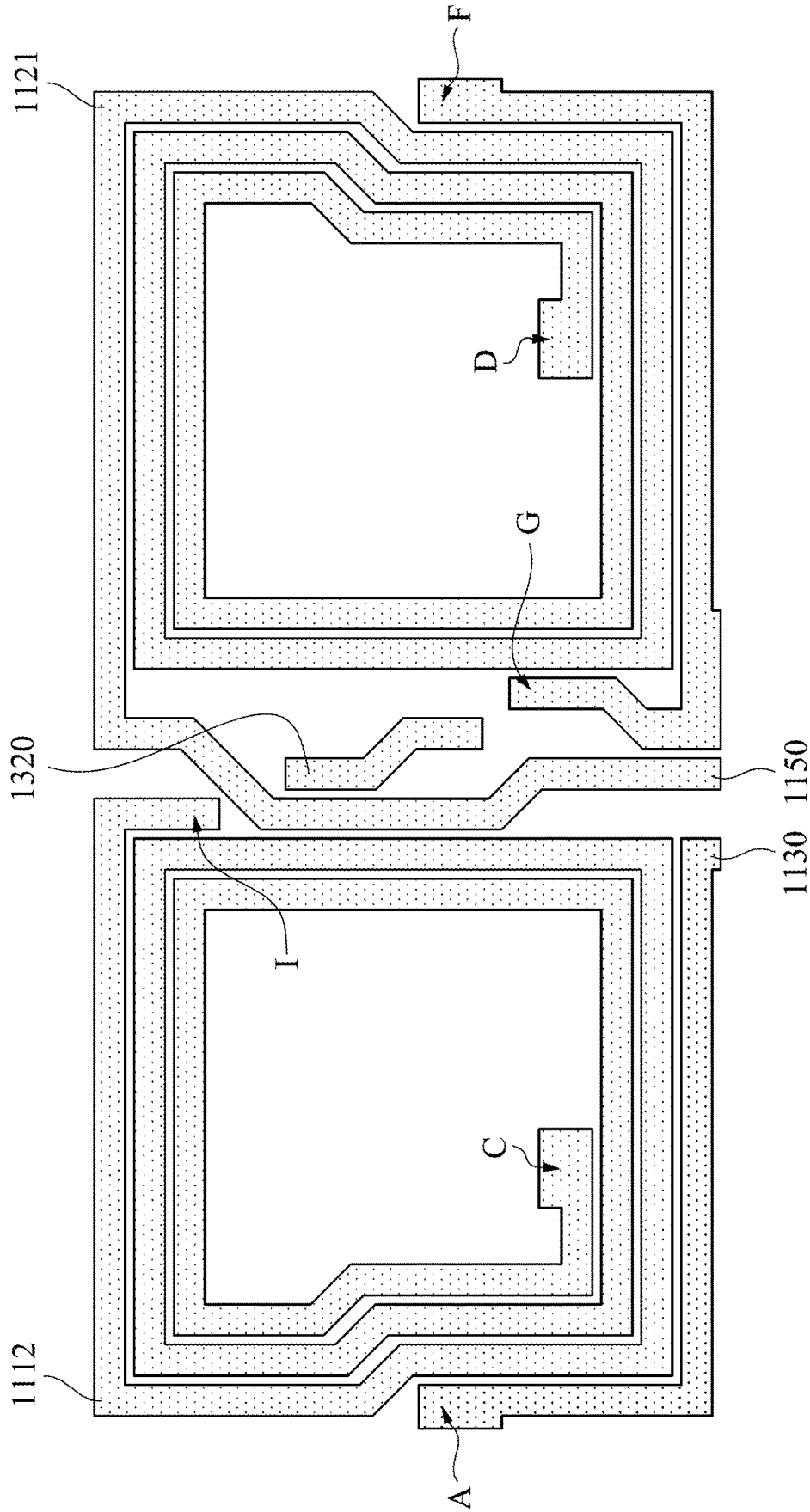
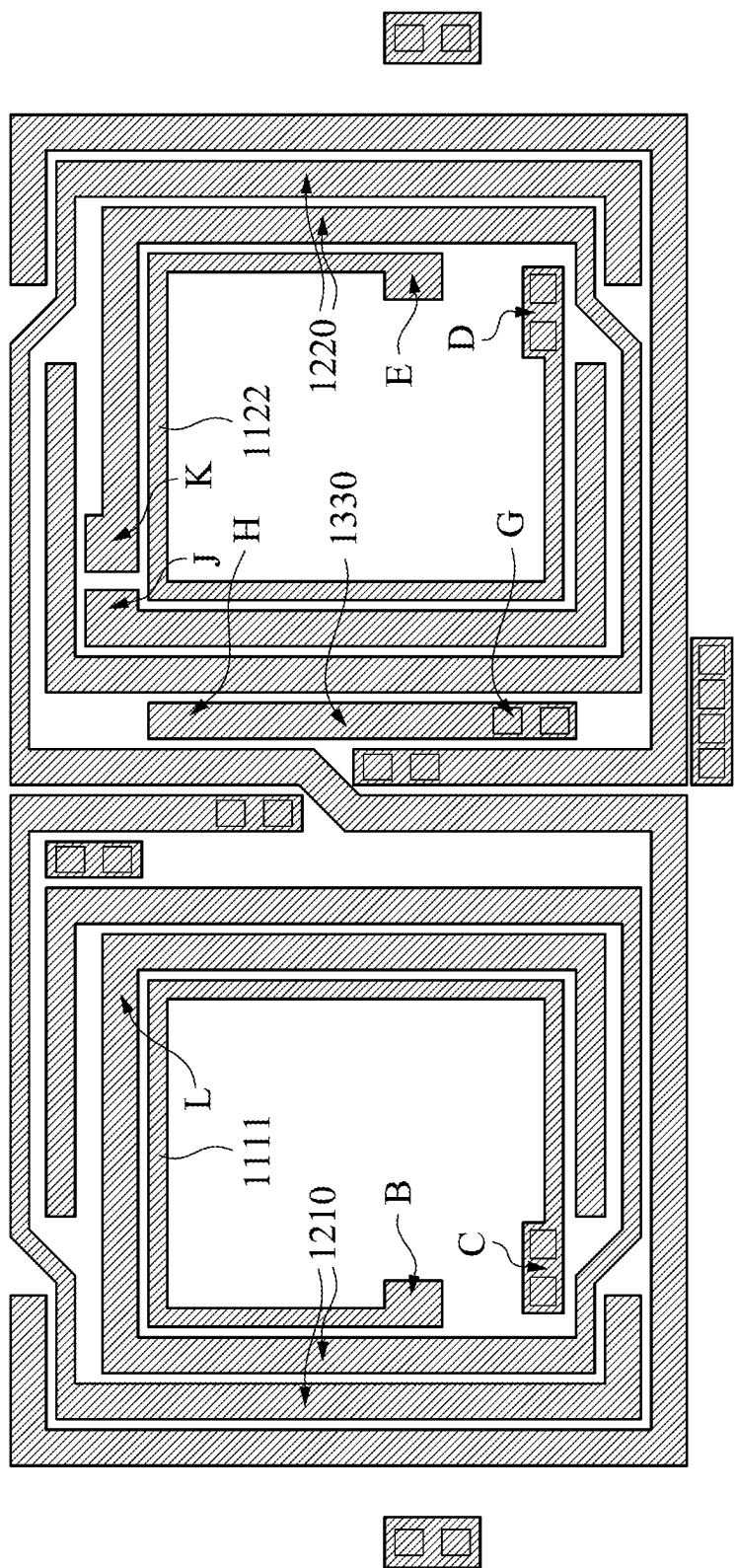


Fig. 2



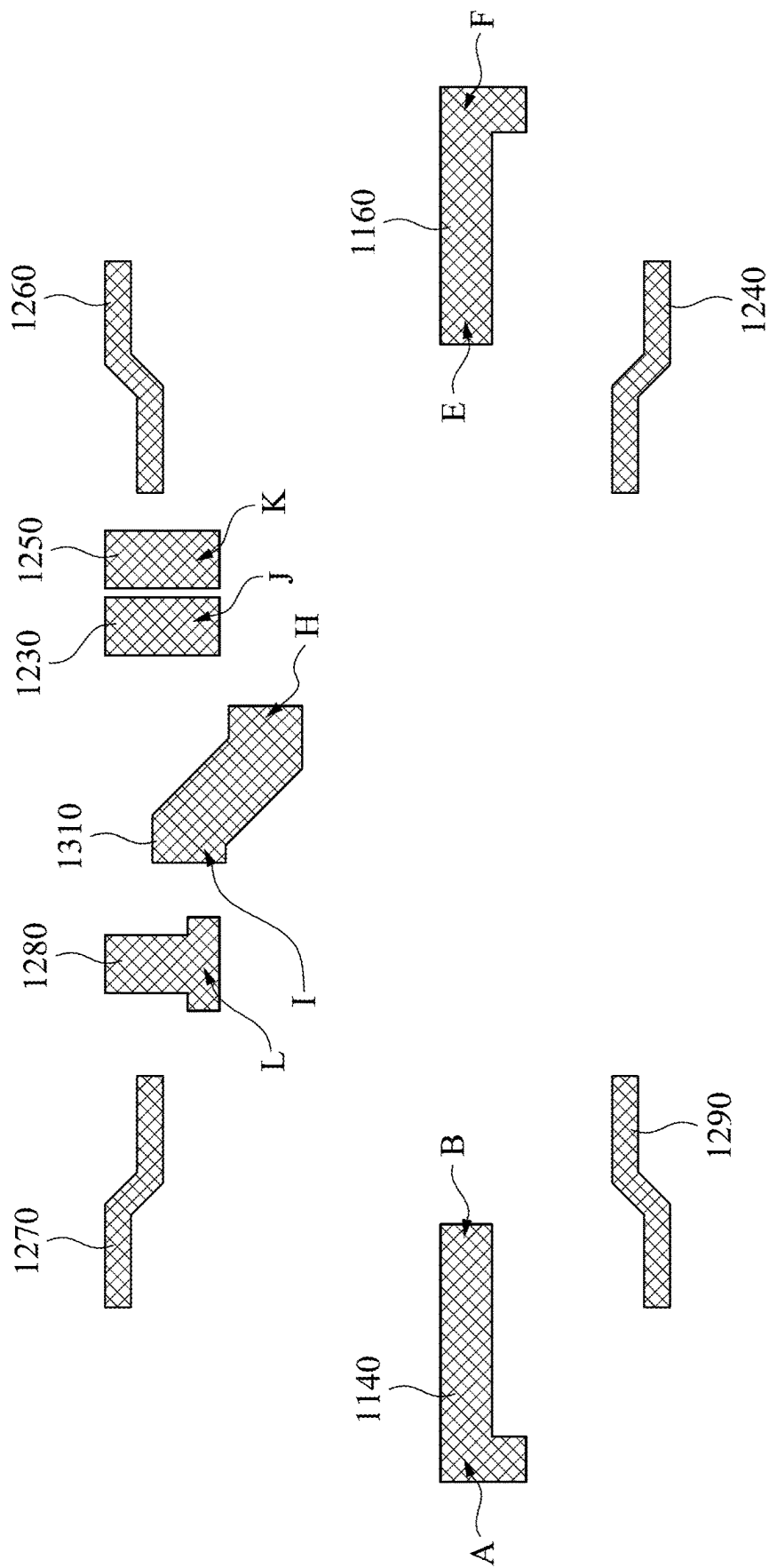


Fig. 4

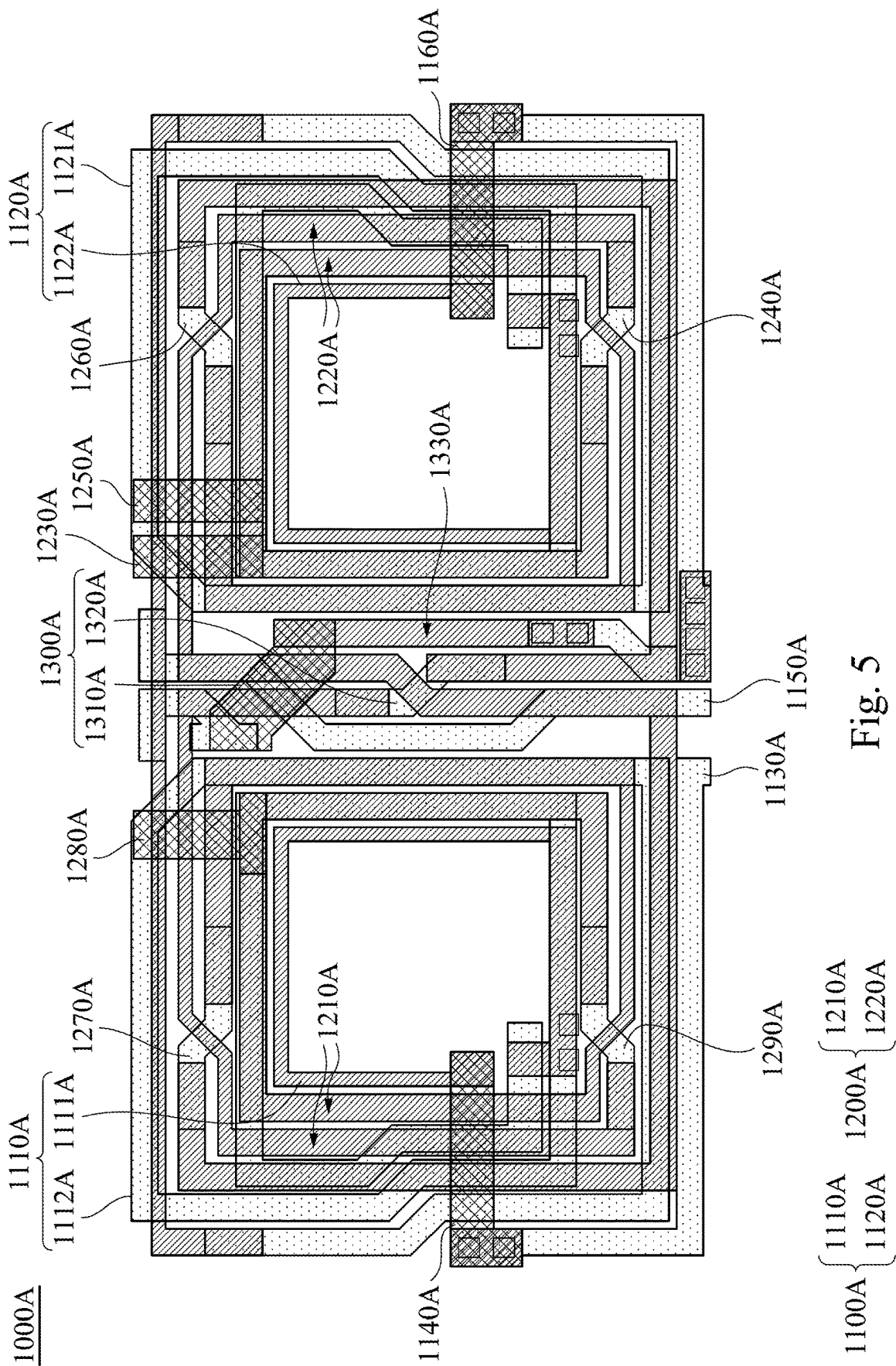


Fig. 5

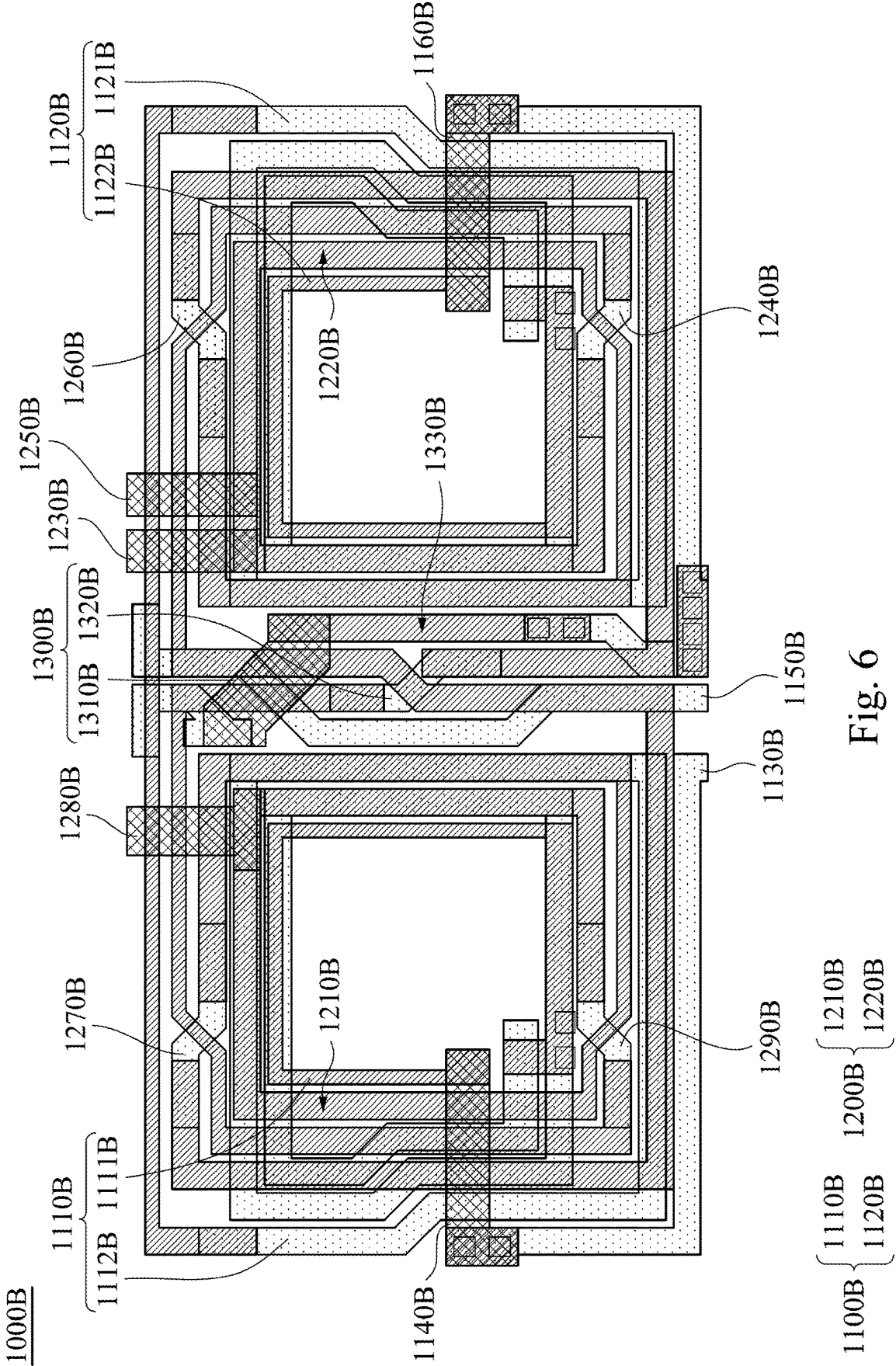


Fig. 6

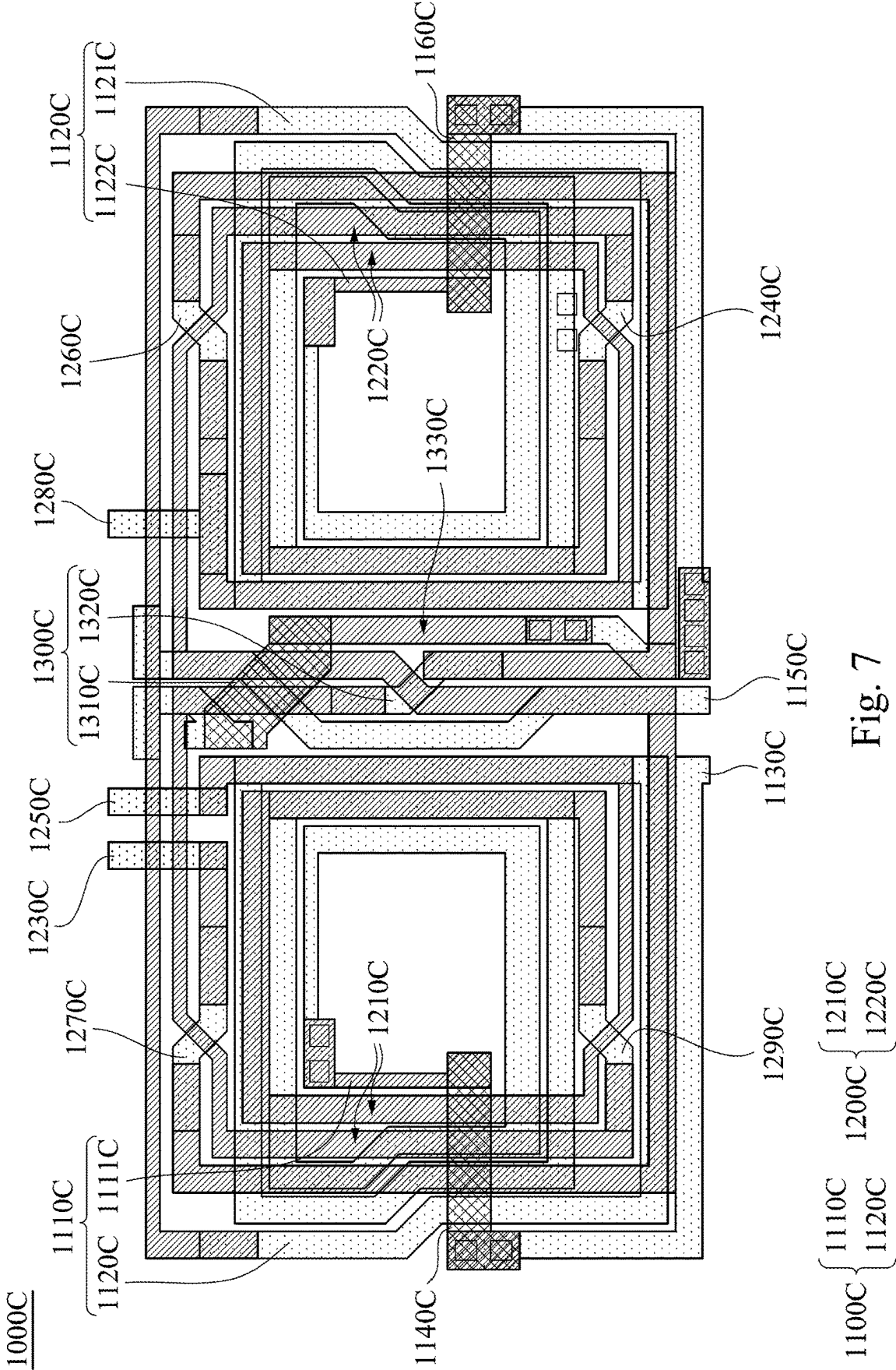


Fig. 7

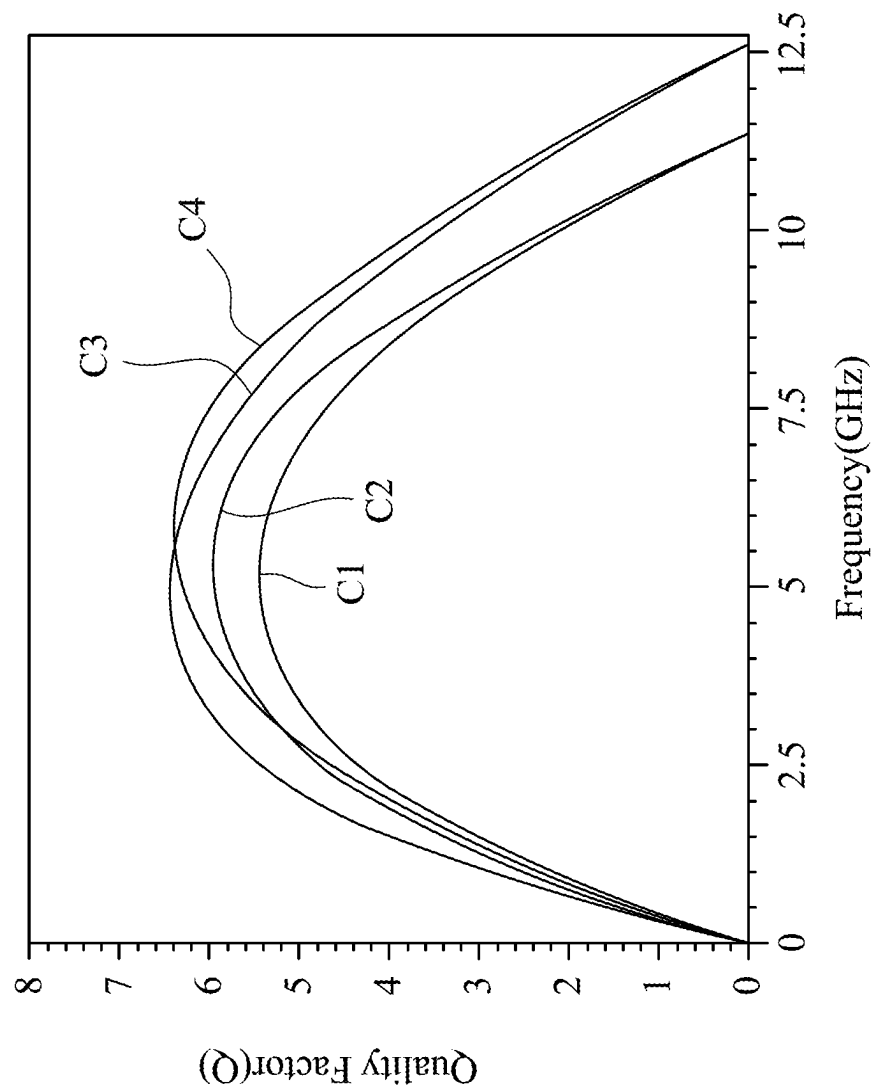


Fig. 8

1

INDUCTOR DEVICE**RELATED APPLICATIONS**

This application claims priority to and the benefit of 5
Taiwan Application Serial Number 110130908, filed on
Aug. 20, 2021, the entire contents of which are incorporated
herein by reference as if fully set forth below in its entirety
and for all applicable purposes.

BACKGROUND**Field of Invention**

The present disclosure relates to an electronic device. 15
More particularly, the present disclosure relates to an inductor
device.

Description of Related Art

The various types of inductors according to the prior art
have their advantages and disadvantages. For example, a
spiral inductor has a higher Q value and a larger mutual
inductance. However, coupling is occurred between a spiral
inductor and other devices. For an eight-shaped inductor/ 25
transformer which has two sets of coils, the coupling
between the two sets of coils is relatively low. However, an
eight-shaped inductor/transformer occupies a larger area in
a device. In addition, when an eight-shaped inductor/trans-
former is designed to be a symmetrical structure, parasitic
capacitances of differential signals of the eight-shaped
inductor/transformer increase substantially due to traditional
crossing structures. Therefore, the scopes of application of
the above inductor/transformer are limited.

SUMMARY

The foregoing presents a simplified summary of the
disclosure in order to provide a basic understanding to the
reader. This summary is not an extensive overview of the
disclosure and it does not identify key/critical elements of
the present disclosure or delineate the scope of the present
disclosure. Its sole purpose is to present some concepts
disclosed herein in a simplified form as a prelude to the more
detailed description that is presented later.

One aspect of the present disclosure is to provide an
inductor device. The inductor device comprises a first induc-
tor, a second inductor, and a connection portion. The first
inductor comprises a first trace and a second trace. The
second inductor comprises a third trace and a fourth trace. A 50
shape of one of the first trace and the second trace comprises
a spiral. The third trace and the fourth trace are symmetrical
to each other. The connection portion is configured to couple
to the first inductor and the second inductor.

Therefore, based on the technical content of the present 55
disclosure, the parasitic capacitance can be decrease owing
to the disposition of the inductor device of the present
disclosure. The differential signals of the symmetrical circuit
are inputted at the same side, for example, the positive
voltage signal or the negative voltage signal (P or N) are
inputted at the same side. A new design, which adopts the
positive voltage signals which are adjacent to each other (P
and P are adjacent to each other), can avoid the capacitance
generated by the positive voltage signal and the negative
voltage signal which are adjacent to each other (N/P are
adjacent to each other) in a traditional manner. Therefore,
the parasitic capacitance decreases owing to the disposition 65

2

of the inductor device of the present disclosure, such that the
quality factor (Q factor) enhances and the summarization of
the structure increases.

It is to be understood that both the foregoing general
description and the following detailed description are by
examples, and are intended to provide further explanation of
the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a
further understanding of the invention, and are incorporated
in and constitute a part of this specification. The drawings
illustrate embodiments of the invention and, together with
the description, serve to explain the principles of the inven- 15
tion. In the drawings,

FIG. 1 depicts a schematic diagram of an inductor device
according to one embodiment of the present disclosure;

FIG. 2 depicts a schematic diagram of a portion structure
of the inductor device shown in FIG. 1 according to one
embodiment of the present disclosure;

FIG. 3 depicts a schematic diagram of a portion structure
of the inductor device shown in FIG. 1 according to one
embodiment of the present disclosure;

FIG. 4 depicts a schematic diagram of a portion structure
of the inductor device shown in FIG. 1 according to one
embodiment of the present disclosure;

FIG. 5 depicts a schematic diagram of an inductor device
according to one embodiment of the present disclosure;

FIG. 6 depicts a schematic diagram of an inductor device
according to one embodiment of the present disclosure;

FIG. 7 depicts a schematic diagram of an inductor device
according to one embodiment of the present disclosure; and

FIG. 8 depicts a schematic diagram of experimental data 35
of an inductor device according to one embodiment of the
present disclosure.

According to the usual mode of operation, various fea-
tures and elements in the figures have not been drawn to
scale, which are drawn to the best way to present specific
features and elements related to the disclosure. In addition,
among the different figures, the same or similar element
symbols refer to similar elements/components.

DESCRIPTION OF THE EMBODIMENTS

To make the contents of the present disclosure more
thorough and complete, the following illustrative description
is given with regard to the implementation aspects and
embodiments of the present disclosure, which is not
intended to limit the scope of the present disclosure. The
features of the embodiments and the steps of the method and
their sequences that constitute and implement the embodi-
ments are described. However, other embodiments may be
used to achieve the same or equivalent functions and step
sequences.

Unless otherwise defined herein, scientific and technical
terminologies employed in the present disclosure shall have
the meanings that are commonly understood and used by
one of ordinary skill in the art. Unless otherwise required by
context, it will be understood that singular terms shall
include plural forms of the same and plural terms shall
include the singular. Specifically, as used herein and in the
claims, the singular forms “a” and “an” include the plural
reference unless the context clearly indicates otherwise.

FIG. 1 depicts a schematic diagram of an inductor device
1000 according to one embodiment of the present disclo-
sure. As shown in the figure, the inductor device 1000

3

includes a first inductor **1100**, a second inductor **1200**, and a connection portion **1300**. The first inductor **1100** includes a first trace **1110** and a second trace **1120**. The second inductor **1200** includes a third trace **1210** and a fourth trace **1220**. As shown in the figure, the connection portion **1300**, which is located on a central area, can be configured to couple to the first inductor **1100** and the second inductor **1200**.

For facilitating the understanding of the inductor device **1000** in FIG. 1, reference is now made to both FIG. 2 to FIG. 4. FIG. 2 to FIG. 4 depict schematic diagrams of a portion structure of the inductor device **1000** shown in FIG. 1 according to one embodiment of the present disclosure. As shown in FIG. 2, in one embodiment, the shape of one of the portion trace **1112** of the first trace **1110** and the portion trace **1121** of the second trace **1120** can be spiral. For example, the shape of one of the portion trace **1112** and the portion trace **1121** can be spiral, or the shapes of both of the portion trace **1112** and the portion trace **1121** are spiral.

As shown in FIG. 3, in one embodiment, the third trace **1210** and the fourth trace **1220** are symmetrical to each other. Specifically, the third trace **1210** and the fourth trace **1220** are symmetrical to each other on the basis of the central area of the inductor device **1000**.

In one embodiment, the portion structure of the inductor device **1000** shown in FIG. 2 is located on a first layer, the portion structure of the inductor device **1000** shown in FIG. 3 is located on a second layer, and the portion structure of the inductor device **1000** shown in FIG. 4 is located on a third layer. For example, referring to FIG. 2, the portion trace **1112** of the first trace **1110**, the portion trace **1121** of the second trace **1120**, and the portion connection portion **1320** of the connection portion **1300** are located on the first layer.

In addition, referring to FIG. 3, the portion trace **1111** of the first trace **1110**, the portion trace **1122** of the second trace **1120**, the third trace **1210**, and the fourth trace **1220** are located on the second layer. Besides, referring to FIG. 4, the first sub-connection portion **1310** of the connection portion **1300** is located on the third layer. Furthermore, the third layer shown in FIG. 4 further includes a plurality of connection members **1140**, **1160**, **1240**, **1260**, **1270**, **1290**, input/output terminals **1230**, **1250**, and a center-tapped terminal **1280**. The above-mentioned structures will be described in detail as shown below. As described above, all structures of the inductor device **1000** of the present disclosure are located on three layers. Compared with the traditional inductor device adopting four layers for including all structures, the inductor device **1000** of the present disclosure can save area, volume, and so on, and the inductor device **1000** of the present disclosure merely needs three layers for including all structures, such that the design of the circuit becomes easier.

In one embodiment, the first trace **1110** can be the first spiral trace **1110**, and the first spiral trace **1110** is disposed on the first layer and the second layer. For example, the portion trace **1112** of the first spiral trace **1110** is disposed on the first layer shown in FIG. 2, and the portion trace **1111** of the first spiral trace **1110** is disposed on the second layer shown in FIG. 3.

In another embodiment, the first inductor **1100** further includes a first input/output terminal **1130** and a first connection member **1140**. The first input/output terminal **1130** is disposed on the first layer shown in FIG. 2, and the first connection member **1140** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the node A which is disposed on the first input/output

4

terminal **1130** shown in FIG. 2 is coupled to the node A which is disposed on the first connection member **1140** shown in FIG. 4. Subsequently, the node B of the first connection member **1140** is coupled to the node B which is disposed on the portion trace **1111** of the first spiral trace **1110** shown in FIG. 3. After that, the node C of the portion trace **1111** of the first spiral trace **1110** is coupled to the node C which is disposed on the portion trace **1112** of the first spiral trace **1110** shown in FIG. 2.

In one embodiment, the second trace **1120** can be the second spiral trace **1120**, and the second spiral trace **1120** is disposed on the first layer and the second layer. For example, the portion trace **1121** of the second spiral trace **1120** is disposed on the first layer shown in FIG. 2, and the portion trace **1122** of the second spiral trace **1120** is disposed on the second layer shown in FIG. 3.

In another embodiment, the first inductor **1100** further includes a second input/output terminal **1150** and a second connection member **1160**. The second input/output terminal **1150** is disposed on the first layer shown in FIG. 2, and the second connection member **1160** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the second input/output terminal **1150** shown in FIG. 2 is coupled to the portion trace **1121** of the second spiral trace **1120** which is also shown in FIG. 2. The node D of the portion trace **1121** of the second spiral trace **1120** is coupled to the node D which is disposed on the portion trace **1122** of the second spiral trace **1120** shown in FIG. 3. Subsequently, the node E of the portion trace **1122** of the second spiral trace **1120** is coupled to the node E which is disposed on the second connection member **1160** shown in FIG. 4.

Next, the node F of the second connection member **1160** is coupled to the node F which is disposed on the portion trace **1121** of the second spiral trace **1120** shown in FIG. 2. The node G of the portion trace **1121** of the second spiral trace **1120** is coupled to the node G of the connection member **1330** shown in FIG. 3. Subsequently, the node H of the connection member **1330** is coupled to the node H which is disposed on the first sub-connection portion **1310** of the connection portion **1300** shown in FIG. 4. The node I of the first sub-connection portion **1310** of the connection portion **1300** is coupled to the node I which is disposed on the portion trace **1112** of the first spiral trace **1110** shown in FIG. 2.

In one embodiment, the second inductor **1200** further includes a third input/output terminal **1230** and a third connection member **1240**. The third input/output terminal **1230** is disposed on the third layer shown in FIG. 4, and the third connection member **1240** is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the node J which is disposed on the third input/output terminal **1230** shown in FIG. 4 is coupled to the node J which is disposed on the fourth trace **1220** shown in FIG. 3. The fourth trace **1220**, which is located on the second layer, is coupled through the third connection member **1240** to the fourth trace **1220**, which is also located on the second layer, in an interlaced manner.

In another embodiment, referring to FIG. 1, the portion trace **1121** of the second spiral trace **1120** located on the first layer and the fourth trace **1220** located on the second layer are overlapped to each other partially.

In one embodiment, the second inductor **1200** further includes a fourth input/output terminal **1250** and a fourth connection member **1260**. The fourth input/output terminal **1250** is disposed on the third layer shown in FIG. 4, and the fourth connection member **1260** is disposed on the third

5

layer shown in FIG. 4. With respect to the connection structure, the node K which is disposed on the fourth input/output terminal 1250 shown in FIG. 4 is coupled to the node K which is disposed on the fourth trace 1220 shown in FIG. 3. The fourth trace 1220 is coupled through the fourth connection member 1260 to the fourth trace 1220 which is also located on the second layer in an interlaced manner. In another embodiment, the third connection member 1240 and the fourth connection member 1260 are located at different sides of the second inductor 1200. For example, the third connection member 1240 and the fourth connection member 1260 are located at the lower side and the upper side of the second inductor 1200 respectively.

In another embodiment, in the central area (e.g. the center of the structure shown in the figure) of the inductor device 1000, the third trace 1210 which is disposed on the second layer shown in FIG. 3 is coupled to the fourth trace 1220 which is also located on the second layer.

In one embodiment, the second inductor 1200 further includes a fifth connection member 1270, and the fifth connection member 1270 is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the third trace 1210 shown in FIG. 3 is coupled through the fifth connection member 1270 shown in FIG. 4 to the third trace 1210 which is also located on the second layer in an interlaced manner.

In another embodiment, referring to FIG. 1, the portion trace 1112 of the first spiral trace 1110 located on the first layer and the third trace 1210 located on the second layer are overlapped to each other partially.

In one embodiment, the second inductor 1200 further includes a center-tapped terminal 1280, and the center-tapped terminal 1280 is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the node L which is disposed on the center-tapped terminal 1280 shown in FIG. 4 is coupled to the node L which is disposed on the third trace 1210 shown in FIG. 3.

In another embodiment, referring to FIG. 1, the center-tapped terminal 1280 is symmetrical to the third input/output terminal 1230 and the fourth input/output terminal 1250 on the basis of the central area (e.g. the center of the structure shown in the figure) of the inductor device 1000.

In one embodiment, the second inductor 1200 further includes a sixth connection member 1290, and the sixth connection member 1290 is disposed on the third layer shown in FIG. 4. With respect to the connection structure, the third trace 1210 in FIG. 3 is coupled through the sixth connection member 1290 shown in FIG. 4 to the third trace 1210 which is also located on the second layer in an interlaced manner. In another embodiment, the fifth connection member 1270 and the sixth connection member 1290 are disposed at different sides of the second inductor 1200. For example, the fifth connection member 1270 and the sixth connection member 1290 are located at the upper side and the lower side of the second inductor 1200 respectively.

In another embodiment, the connection portion 1300 further includes a second sub-connection portion 1320, and the second sub-connection portion 1320 is disposed on the first layer shown in FIG. 2. As shown in FIG. 1, in the central area (e.g. the center of the structure shown in the figure) of the inductor device 1000, the third trace 1210 located on the second layer is coupled through the second sub-connection portion 1320 to the fourth trace 1220 located on the second layer. It is noted that the present disclosure is not limited to the structure as shown in FIGS. 1 to 4, and it is merely an example for illustrating one of the implements of the present disclosure.

6

FIG. 5 depicts a schematic diagram of an inductor device 1000A according to one embodiment of the present disclosure. Compared with the inductor device 1000 shown in FIG. 1, the structure disposition of the inductor device 1000A shown in FIG. 5 is different, which will be described in detail as below.

Referring to FIG. 5, the structure between the first trace 1110A and the third trace 1210A of the inductor device 1000A is adjusted, so as to make the fifth connection member 1270A located at the upper-left corner in the figure to be disposed on the first layer, and make the sixth connection member 1290A located at the lower-left corner in the figure to be disposed on the first layer.

Besides, the structure between the second trace 1120A and the fourth trace 1220A of the inductor device 1000A is adjusted, so as to make the fourth connection member 1260A located at the upper-right corner in the figure to be disposed on the first layer, and make the third connection member 1240A located at the lower-right corner in the figure to be disposed on the first layer. It is noted that the element in FIG. 5, whose symbol is similar to the symbol of the element in FIG. 1, has similar structure feature in connection with the element in FIG. 1. Therefore, a detail description regarding the structure feature of the element in FIG. 5 is omitted herein for the sake of brevity. In addition, the present disclosure is not limited to the structure as shown in FIG. 5, and it is merely an example for illustrating one of the implements of the present disclosure.

FIG. 6 depicts a schematic diagram of an inductor device 1000B according to one embodiment of the present disclosure. Compared with the inductor device 1000 shown in FIG. 1, the structure disposition of the inductor device 1000B shown in FIG. 6 is different, which will be described in detail as below.

Referring to FIG. 6, the structure between the first trace 1110B and the third trace 1210B of the inductor device 1000B is adjusted, so as to make the fifth connection member 1270B located at the upper-left corner in the figure to be disposed on the first layer, and make the sixth connection member 1290B located at the lower-left corner in the figure to be disposed on the first layer.

Besides, the structure between the second trace 1120B and the fourth trace 1220B of the inductor device 1000B is adjusted, so as to make the fourth connection member 1260B located at the upper-right corner in the figure to be disposed on the first layer, and make the third connection member 1240B located at the lower-right corner in the figure to be disposed on the first layer. It is noted that the element in FIG. 6, whose symbol is similar to the symbol of the element in FIG. 1, has similar structure feature in connection with the element in FIG. 1. Therefore, a detail description regarding the structure feature of the element in FIG. 6 is omitted herein for the sake of brevity. In addition, the present disclosure is not limited to the structure as shown in FIG. 6, and it is merely an example for illustrating one of the implements of the present disclosure.

FIG. 7 depicts a schematic diagram of an inductor device 1000C according to one embodiment of the present disclosure. Compared with the inductor device 1000 shown in FIG. 1, the structure disposition of the inductor device 1000C shown in FIG. 7 is different, which will be described in detail as below.

Referring to FIG. 7, the structure between the first trace 1110C and the third trace 1210C of the inductor device 1000C is adjusted, so as to make the fifth connection member 1270C located at the upper-left corner in the figure to be disposed on the first layer, and make the sixth con-

7

nection member **1290C** located at the lower-left corner in the figure to be disposed on the first layer.

Besides, the structure between the second trace **1120C** and the fourth trace **1220C** of the inductor device **1000C** is adjusted, so as to make the fourth connection member **1260C** located at the upper-right corner in the figure to be disposed on the first layer, and make the third connection member **1240C** located at the lower-right corner in the figure to be disposed on the first layer.

In addition, the input/output terminals **1230C**, **1250C** of the inductor device **1000C** can be disposed on the first layer, and the input/output terminals **1230C**, **1250C** are coupled to the third sub-trace **1210C**. Furthermore, the center-tapped terminal **1280C** of the inductor device **1000C** can be disposed on the first layer, and the center-tapped terminal **1280C** is coupled to the fourth sub-trace **1220C**. It is noted that the element in FIG. 7, whose symbol is similar to the symbol of the element in FIG. 1, has similar structure feature in connection with the element in FIG. 1. Therefore, a detail description regarding the structure feature of the element in FIG. 7 is omitted herein for the sake of brevity. In addition, the present disclosure is not limited to the structure as shown in FIG. 7, and it is merely an example for illustrating one of the implements of the present disclosure.

FIG. 8 depicts a schematic diagram of experimental data of an inductor device according to one embodiment of the present disclosure. As shown in the figure, the experimental curves of the quality factors of the inductor device not adopting the structural configuration of the present disclosure are C1, C2, and the experimental curves of the quality factors of the inductor device adopting the structural configuration of the present disclosure are C3, C4. As shown in the figure, the inductor device adopting the structural configuration of the present disclosure has better the quality factors.

Although the present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims and their equivalents.

What is claimed is:

1. An inductor device, comprising:

a first inductor, comprising:

a first trace, wherein the first trace comprises a first spiral trace, wherein the first spiral trace is disposed on a first layer and a second layer;

a second trace, wherein the second trace comprises a second spiral trace, wherein the second spiral trace is disposed on the first layer and the second layer; and

a second connection member, disposed on a third layer, wherein the second spiral trace which is located on the second layer is coupled to the second spiral trace which is located on the first layer through the second connection member;

a second inductor, comprising:

a third trace; and

a fourth trace, wherein the third trace and the fourth trace are symmetrical to each other; and

a connection portion, configured to couple to the first inductor and the second inductor, wherein the connec-

8

tion portion comprises a first sub-connection portion, configured to couple to the second spiral trace which is located on the first layer and the first spiral trace which is located on the first layer.

2. The inductor device of claim 1, wherein the first inductor is located on the first layer and the second layer, and the second inductor is located on the second layer, and the connection portion is located on the first layer and the third layer.

3. The inductor device of claim 1, wherein the first inductor further comprises:

a first input/output terminal, disposed on the first layer, and coupled to the first spiral trace which is located on the second layer, wherein the first spiral trace which is located on the second layer is coupled to the first spiral trace which is located on the first layer.

4. The inductor device of claim 3, wherein the first inductor further comprises:

a first connection member, disposed on the third layer, wherein the first input/output terminal is coupled to the first spiral trace which is located on the second layer through the first connection member.

5. The inductor device of claim 1, wherein the first inductor further comprises:

a second input/output terminal, disposed on the first layer, and coupled to the second spiral trace which is located on the first layer, wherein the second spiral trace which is located on the first layer is coupled to the second spiral trace which is located on the second layer.

6. The inductor device of claim 1, wherein the second spiral trace which is located on the first layer and the fourth trace which is located on the second layer are overlapped to each other partially.

7. The inductor device of claim 1, wherein the first spiral trace which is located on the first layer and the third trace which is located on the second layer are overlapped to each other partially.

8. The inductor device of claim 1, wherein the second inductor further comprises:

a third input/output terminal, disposed on the third layer, and coupled to the fourth trace which is located on the second layer.

9. The inductor device of claim 8, wherein the second inductor further comprises:

a fourth input/output terminal, disposed on the third layer, and coupled to the fourth trace which is located on the second layer.

10. The inductor device of claim 9, wherein in a central area of the inductor device, the third trace which is located on the second layer is coupled to the fourth trace which is located on the second layer.

11. The inductor device of claim 10, wherein the second inductor further comprises:

a center-tapped terminal, disposed on the third layer, and coupled to the third trace which is located on the second layer.

12. The inductor device of claim 11, wherein the center-tapped terminal is disposed symmetrically to the third input/output terminal and the fourth input/output terminal on a basis of the central area.

13. The inductor device of claim 10, wherein the connection portion further comprises:

a second sub-connection portion, disposed on the first layer, wherein in the central area of the inductor device, the third trace, which is located on the second layer, is coupled through the second sub-connection portion to the fourth trace, which is located on the second layer.

14. The inductor device of claim **1**, wherein the second inductor further comprises:

a third connection member, disposed on the third layer, wherein the fourth trace, which is located on the second layer, is coupled through the third connection member to the fourth trace, which is located on the second layer, in an interlaced manner. 5

15. The inductor device of claim **14**, wherein the second inductor further comprises:

a fourth connection member, disposed on the third layer, wherein the fourth trace, which is located on the second layer, is coupled through the fourth connection member to the fourth trace, which is located on the second layer, in an interlaced manner, wherein the third connection member and the fourth connection member are located at different sides of the second inductor. 10 15

16. The inductor device of claim **1**, wherein the second inductor further comprises:

a fifth connection member, disposed on the third layer, wherein the third trace, which is located on the second layer, is coupled through the fifth connection member to the third trace, which is located on the second layer, in an interlaced manner. 20

17. The inductor device of claim **16**, wherein the second inductor further comprises: 25

a sixth connection member, disposed on the third layer, wherein the third trace, which is located on the second layer, is coupled through the sixth connection member to the third trace, which is located on the second layer, in an interlaced manner, wherein the fifth connection member and the sixth connection member are located at different sides of the second inductor. 30

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